

Complete Learning Material in Q&A Format

A. Basic Ideas and Plate Tectonics

Q1. Why does the Earth's surface keep changing?

The Earth's surface keeps changing because powerful forces act **inside the Earth** and **on its surface**. Plate movement, earthquakes and volcanic activity act from inside, while weathering, erosion and deposition reshape the surface over time. These processes create mountains, valleys, plains, volcanoes and many other landforms.

Q2. What is a landform? Give examples.

A **landform** is a natural feature of the Earth's surface. It may be formed by weathering, erosion, deposition or movement of the Earth's crust. Mountains, valleys, plateaus, plains, deserts and coastal features are examples.

Q3. What is plate tectonics?

Plate tectonics is the theory that explains the movement of the Earth's crust. According to the chapter, this theory was given by **W.J. Morgan**. The outer part of the Earth is broken into large and small pieces called **tectonic plates**. These plates move slowly over a semi-molten [partly melted] layer below them.

Q4. Why is plate tectonics important?

Plate tectonics helps us understand:

- formation of mountains and valleys,
- formation of continents and ocean basins,
- earthquakes,
- volcanoes, and
- continuous changes in the Earth's surface.

It can also help identify areas that are more likely to experience earthquakes and volcanic activity.

B. Interior of the Earth

Q5. What are the three main layers of the Earth?

The three main layers are:

1. **Crust** – the outermost layer on which we live.
2. **Mantle** – a very thick and hot layer below the crust.
3. **Core** – the innermost, extremely hot and heavy part of the Earth.

Q6. What is the crust?

The crust is the Earth's outermost layer. Its thickness is about **30–40 km under continents** and about **5–7 km under oceans**.

Q7. What is the mantle?

The mantle is the mostly solid layer between the crust and the Earth's outer core. It is very thick and hot. The diagram gives its approximate thickness as **2,900 km**.

Q8. What is the core?

The core is the innermost part of the Earth. It has two main parts:

- **Outer core:** a fluid [liquid-like] layer mainly made of iron and nickel.
- **Inner core:** a solid, very hot and dense [closely packed and heavy] metallic part.

Q9. What is the lithosphere?

The **lithosphere** is the rigid [hard and solid] outer layer of the Earth. It includes the crust and the upper part of the mantle. It is broken into tectonic plates.

Q10. What is the asthenosphere?

The **asthenosphere** is a hot, mobile and partly molten layer below the lithosphere. The tectonic plates can move over this layer.

Q11. What is the difference between the lithosphere and asthenosphere?

Lithosphere	Asthenosphere
Rigid outer layer	Hot and mobile layer
Includes crust and upper mantle	Lies below the lithosphere
Broken into tectonic plates	Allows plates to move
Mostly solid and rigid	Partly molten

C. Tectonic Plates and Their Movement**Q12. What are tectonic plates?**

Tectonic plates are huge slabs [large pieces] of solid rock that form the lithosphere. They move very slowly, usually only a **few centimetres each year**.

Q13. What are the three types of tectonic plates?

They are:

- **Continental plates** – carry continents.
- **Oceanic plates** – carry ocean floors.
- **Mixed plates** – carry both continents and oceans.

Q14. Name the major tectonic plates mentioned in the chapter.

They are:

- Pacific Plate
- Eurasian Plate
- African Plate
- North American Plate
- South American Plate
- Indo-Australian Plate
- Antarctic Plate

Q15. What causes tectonic plates to move?

Tectonic plates move because of **convection currents in the mantle**. Heat from the Earth's core makes hotter material rise. Cooler material sinks. This continuous circulation pushes and pulls the plates.

Q16. What is a convection current?

A **convection current** is a circular movement caused when hotter material rises and cooler material sinks. In the mantle, these currents help move tectonic plates.

Q17. What is a plate boundary?

A **plate boundary** is the edge or area where two tectonic plates meet.

Q18. What are the three main types of plate boundaries?

They are:

1. Convergent boundary
 2. Divergent boundary
 3. Transform boundary
-

D. Types of Plate Boundaries

Q19. What happens at a convergent boundary?

At a **convergent boundary**, two plates move **towards each other**.

Q20. What happens when two continental plates collide?

When two continental plates collide, the crust is pushed and folded. This can form **fold mountains**. The **Himalaya** is given as an example in the chapter.

Q21. What happens when an oceanic plate collides with a continental plate?

The oceanic plate sinks below the continental plate. This process may lead to **earthquakes and volcanic activity**.

Q22. What happens at a divergent boundary?

At a **divergent boundary**, two plates move away from each other. Magma [hot molten rock] rises from below and forms new crust. This may create **mid-ocean ridges**.

Q23. Give an example of a divergent boundary landform.

The **Mid-Atlantic Ridge** is given as an example.

Q24. What happens at a transform boundary?

At a **transform boundary**, two plates slide past each other. No new crust is formed and no crust is destroyed. This movement mainly causes earthquakes.

Q25. Which example of a transform boundary is given in the chapter?

The **San Andreas Fault in the United States** is given as an example.

Q26. Compare the three types of plate boundaries.

Boundary	Movement	Main Result
Convergent	Plates move towards each other	Fold mountains, earthquakes, volcanoes
Divergent	Plates move apart	New crust and mid-ocean ridges
Transform	Plates slide past each other	Mainly earthquakes

E. Earthquakes, Volcanoes and the Ring of Fire

Q27. Why do many earthquakes and volcanoes occur near plate boundaries?

Plate boundaries are places where plates collide, separate or slide past each other. These movements disturb the crust. Therefore, most earthquakes and volcanoes are concentrated near plate boundaries.

Q28. What is the Ring of Fire?

The **Ring of Fire** is the large area around the **Pacific Ocean** where many earthquakes and volcanoes occur. It is closely connected with plate boundaries.

Q29. What correlation [connection] can we see between plate boundaries and the distribution of earthquakes and volcanoes?

When the plate map is compared with the map of earthquakes and volcanoes, many earthquake and volcanic zones are found close to plate boundaries. This shows that plate movement is strongly connected with these events.

Q30. Is India at risk from earthquakes?

Yes. The chapter states that India has experienced major earthquakes that caused thousands of deaths. A large earthquake in a densely populated country can seriously damage people, property and the environment. The chapter shows the **2001 Gujarat earthquake** as an example.

Q31. What was an ancient Indian name for an earthquake?

Earthquakes were known as **bhūkampa**, meaning “shaking of the Earth.”

Q32. What did Varāhamihira write about earthquakes?

In the **Bṛihatsamhitā**, Varāhamihira discussed earthquakes. He noted changes in wind, rain, clouds, animal behaviour and planetary alignments. He linked earthquakes with four elemental forces—**Vāyu**, **Agni**, **Indra** and **Varuṇa**. The chapter presents this as an early attempt to combine observations with cosmological [ideas about the universe] reasoning.

Q33. What does the photograph of grey volcanic material in the chapter show?

It shows **material deposited by a volcanic eruption**. It demonstrates that powerful internal forces of the Earth can bring material to the surface and change the landscape.

Q34. What is special about the mud volcano at Baratang Island?

The mud volcano at **Baratang Island in the Andaman and Nicobar Islands** is a rare natural feature. Unlike fiery volcanoes, mud bubbles out because of underground gases and pressure.

F. Weathering

Q35. What is weathering?

Weathering is the breaking of rocks on the Earth's surface into smaller pieces. The broken material **does not move away** during weathering.

Q36. What are the three types of weathering?

They are:

1. Physical weathering
2. Chemical weathering
3. Biological weathering

Q37. What is physical weathering?

Physical weathering happens when rocks break into smaller pieces because of factors such as:

- heating and cooling,
- temperature changes,
- frost,
- wind,
- water, and
- expansion of ice.

The basic material of the rock is not described as chemically changing.

Q38. What is chemical weathering?

Chemical weathering happens when minerals in rocks react with **water, air or acids** and change into new substances.

Q39. What is biological weathering?

Biological weathering is caused by plants, animals or micro-organisms. For example, plant roots may grow into cracks in a rock and gradually split it apart.

Q40. Why is weathering important?

Weathering helps:

- shape the Earth's surface, and
 - form soil.
-

G. Erosion

Q41. What is erosion?

Erosion is the wearing away **and movement** of soil, rocks and other surface materials from one place to another by natural agents such as water, wind, ice and waves.

Q42. What is the main difference between weathering and erosion?

Weathering

Breaks rocks into smaller pieces

Erosion

Wears away and carries material

Weathering

Broken material does not move

May be physical, chemical or biological

Erosion

Material moves to another place

May be caused by water, wind, glaciers or waves

Q43. What are the main types of erosion mentioned in the chapter?

They are:

- **Water erosion** – caused by rivers, rain or ocean waves.
- **Wind erosion** – common in dry and sandy areas.
- **Glacial erosion** – moving ice scrapes and carries rocks.
- **Coastal erosion** – sea waves wear away coastal land.

Q44. How do weathering and erosion work together to shape the Earth?

Weathering first breaks rocks into smaller pieces. Erosion can then remove and carry these materials. Over a long period, the two processes can wear down mountains, carve valleys, form plains and create features such as caves, cliffs and river deltas.

H. Effects of Erosion on Human Life

Q45. How does erosion affect farmers?

Erosion removes fertile **topsoil** [upper fertile soil layer]. This reduces soil quality and may lower crop production.

Q46. How can erosion affect people living near rivers and coasts?

It can wash away:

- land,
- houses,
- roads, and
- other property.

This can cause serious damage and economic loss.

Q47. How does erosion affect construction, mining, tourism and fishing?

Erosion can make land unstable, creating danger in construction and mining. It may also damage beaches, river areas and fertile lands, affecting tourism and fishing.

Q48. What traditional methods of soil and water management are mentioned in the chapter?

The chapter mentions:

- contouring,
- bunding,
- terracing,
- dams,
- canals, and
- check dams.

The Sindhu-Sarasvatī civilisation used several such techniques. Ancient texts including the **Vedas**, **Kṛīṣhiparāśhara**, **Arthaśhāstra** and **Vṛikṣhāyurveda** also mention land and water management practices.

Q49. What is contouring?

In a continuous contour trench, trenches are dug along the contour lines [lines joining places of equal height] of a hillside. They slow and hold rainwater, reduce soil erosion and help recharge groundwater.

Q50. What is bunding?

Bunding means building earthen embankments [raised barriers of soil] along contour lines. These slow

surface run-off, reduce soil erosion and increase water infiltration [water entering the soil] and soil moisture.

Q51. What is terracing?

Terracing means creating a series of level or gently sloping steps on a hillside. It helps prevent soil erosion.

Q52. What is the Zabo system?

The **Zabo system of Nagaland** is an integrated farming method using earthen bunds on hill slopes for soil and water conservation. Check dams across small streams reduce the speed of water, prevent erosion, allow sediments to settle and help groundwater recharge.

Agents of Gradation

Q53. What are agents of gradation?

Agents of gradation are natural forces that:

- wear down material,
- transport it, and
- deposit it elsewhere.

Over time, they help lower high areas and fill low areas.

Q54. Name the main agents of gradation.

The main agents are:

1. Running water
2. Glaciers
3. Wind
4. Waves
5. Groundwater

Q55. How does each major agent of gradation change the land?

Running water creates valleys and plains. Glaciers carve U-shaped valleys. Wind erodes and deposits sand in deserts. Sea waves shape cliffs, beaches and bays. Groundwater dissolves rocks such as limestone and creates caves and sinkholes.

Running Water

Q56. How do rivers shape the land?

Rivers shape land by three main processes:

- **erosion** – wearing away,
- **transportation** – carrying materials,
- **deposition** – dropping materials.

Different landforms are created in different parts of a river's course.

Q57. Which landforms commonly develop in the upper, middle and lower courses of a river?

River course Common landforms

Upper course V-shaped valleys, waterfalls, rapids

Middle course Meanders, oxbow lakes, floodplains

Lower course Deltas, levees, alluvial fans

I. Waterfalls

Q58. What is a waterfall and how does it form?

A waterfall is a place where a river flows over a steep cliff or vertical drop. It commonly forms in the upper course. Hard rock resists erosion while softer rock below is worn away faster. This creates a sudden drop.

Q59. How are waterfalls useful to humans?

Waterfalls:

- attract tourists,
 - support local economies,
 - may be used for hydroelectric power,
 - provide places for trekking and photography, and
 - may have cultural or religious importance.
-

J. Meanders

Q60. What is a meander? How is it formed?

A **meander** is a winding bend in the middle or lower course of a river. The river erodes its **outer bank** and deposits sediment on the **inner bank**. With time, larger loops develop.

Q61. How are meanders useful to humans?

The deposited soil near meanders is fertile and supports farming. Settlements may develop on gentle slopes nearby. Meandering rivers may also support navigation, irrigation and tourism. The **Grand Anicut or Kallanai in Tamil Nadu** is mentioned as an example of river water being used for irrigation.

K. Deltas

Q62. What is a delta? How is it formed?

A **delta** is a depositional landform formed at the mouth of a river, where the river enters a sea, ocean or lake. The river deposits the sediments it has carried. Over time, the sediment may form a fan-shaped or triangular area of land.

Q63. Why are deltas important to humans?

Deltas often have rich **alluvial soil** [fertile soil deposited by rivers]. They support crops such as rice and jute. They also support fishing, dense settlements, trade and transport. However, delta regions can be vulnerable to flooding.

Waves and Coastal Landforms

Q64. How do waves and currents shape coastal areas?

Waves and currents continuously move along the ocean surface. They erode, transport and deposit materials along coasts. They form beaches, sand bars, sea cliffs, caves, arches and stacks.

Q65. What is a beach?

A beach is an area of sand, pebbles or rocks along the shore of a sea, ocean or lake. It is mainly created by the **deposition of sediments by waves**. Waves, tides and currents continue to change its shape.

Q66. Why are beaches important to humans?

Beaches:

- support tourism and recreation,
- support fishing,
- may provide sand and shells to coastal communities, and
- act as natural barriers against strong waves and coastal erosion.

Q67. What is coastal erosion?

Coastal erosion is the wearing away of coastal land by **waves, tides and currents**.

Q68. Explain the main coastal erosional landforms.

The chapter explains:

- **Sea cliff:** a steep rock face formed as waves undercut [erode underneath] the coast.
 - **Wave-cut platform:** a flat rocky area left when a cliff retreats.
 - **Sea cave:** formed when waves erode a weak part of coastal rock.
 - **Sea arch:** formed when caves cut through a headland.
 - **Sea stack:** an isolated pillar of rock left after an arch collapses.
-

Glaciers

Q69. What is glacial erosion?

Glacial erosion happens when a glacier slowly moves over land and carves or reshapes it.

Q70. Which important landforms are created by glacial erosion?

They include:

- **U-shaped valleys** – river valleys widened and deepened by glaciers.
- **Cirques** – bowl-shaped depressions at the head of a glacier.
- **Aretes** – sharp ridges between valleys.
- **Hanging valleys** – formed where smaller glacial valleys meet larger ones.
- **Fjords** – deep, narrow inlets formed when sea water floods glacial valleys.

Q71. Why are glaciers and glacial landforms important to humans?

U-shaped valleys and cirques attract tourists for trekking, skiing and mountaineering. Fjords may be used as harbours and fishing areas. Some glacial soils support farming. Glaciers are also important sources of fresh water because they feed rivers used by people downstream.

Q72. What is a moraine?

A **moraine** is a deposit of rocks, soil and debris left behind by a glacier. This material is called **till**. When a glacier melts, it deposits the material that it earlier carried.

Q73. What are the three types of moraines mentioned in the chapter?

They are:

- **Lateral moraine:** forms along the sides of a glacier.
- **Terminal moraine:** forms at the end of a glacier and marks its furthest advance.
- **Medial moraine:** forms in the middle when two glaciers meet and their lateral moraines join.

Q74. How are moraines useful to humans?

Moraines may create fertile soil. They can also form natural dams and lakes that may be used for water supply, irrigation and sometimes hydroelectric power.

Q75. What does the chapter tell us about the Chamoli flood of February 2021?

The chapter states that a devastating flood struck **Chamoli district, Uttarakhand, in February 2021**. People and livestock died, while buildings, roads, bridges and hydel projects were badly damaged. Village connectivity was also affected. The textbook asks students to investigate the exact reasons for the sudden flood; it does **not provide those reasons in this section**.

Wind

Q76. How does wind erosion occur?

Strong winds pick up and carry loose sand and soil. Over time, this wears away and reshapes the landscape, especially in dry regions.

Q77. What wind-eroded landforms are mentioned in the chapter?

They include:

- **Yardangs:** streamlined rock ridges carved by wind.
- **Ventifacts:** rocks polished and shaped by wind-driven sand.
- **Deflation hollows or blowouts:** shallow depressions where loose material has been removed.
- **Desert pavements:** flat surfaces left after fine particles are blown away.

Q78. What is a dune?

A **dune** is a hill or ridge of sand formed by wind. Dunes may occur in deserts and along sandy coasts.

Q79. What are the main types of dunes described in the chapter?

The text describes:

- **Barchan dunes:** crescent-shaped; form where sand is limited and wind mainly comes from one direction.
- **Longitudinal dunes:** long ridges running parallel to the prevailing [usual] wind.
- **Star dunes:** have several arms and develop where winds come from different directions.
- **Parabolic dunes:** U-shaped dunes, often stabilised [held in place] by vegetation.

The diagram also labels transverse dunes.

Q80. How are dunes connected with human life?

Dunes may act as barriers against desertification [spread of desert-like conditions] and wind erosion. Coastal dunes may protect settlements from strong sea winds and waves. Dunes also support tourism and adventure activities, and some dune sand is used in construction.

Underground Water and Karst Landforms

Q81. What is Karst topography?

Karst topography is a special landscape produced mainly in limestone or other soluble [easily dissolved] rocks by chemical weathering and erosion caused by underground water.

Q82. How does underground water form caves?

Acidic underground water slowly dissolves soluble rock such as limestone. Over time, hollow spaces develop and become caves.

Q83. What is a stalactite?

A **stalactite** is an icicle-shaped rock formation that hangs **from the ceiling** of a cave.

Q84. What is a stalagmite?

A **stalagmite** is a rock formation that rises **upward from the floor** of a cave.

Q85. What is a sinkhole or doline?

A sinkhole or doline is a depression formed when the ground collapses into an underground cavity [empty space].

Q86. What is an underground river?

An underground river is a river that flows through a cave system below the ground.

Q87. Why are underground-water landforms important to humans?

Caves and underground rivers may provide fresh water and tourism opportunities. Some caves have cultural or religious importance. Stalactites and stalagmites attract geologists [scientists who study the Earth] and adventurers.

Landforms and Human Civilisation

Q88. How have landforms influenced human history and civilisation?

Landforms have strongly influenced settlements, farming, trade, travel, wars and cultural development.

- Fertile river plains of the **Ganga, Nile, Brahmaputra and Indus** supported agriculture and early cities.
 - Mountains could act as barriers and protectors.
 - The Himalayas protected India but mountain passes also allowed cultural contact.
 - Deserts such as the Thar limited large settlements but encouraged trade routes.
 - Coasts and harbours encouraged trade and cultural contact with distant areas.
-

Landforms and Disasters

L. Landslides

Q89. What is a landslide and what causes it?

A landslide occurs when material on a slope becomes unstable and moves downward. Causes mentioned in the chapter include:

- heavy and continuous rainfall,
- earthquakes,
- volcanic eruptions,
- steep slopes,
- loose or weathered rocks,
- deforestation,
- mining,
- road construction,
- unplanned construction,
- poor drainage, and
- improper land use.

Rainwater can enter soil and rocks, increase their weight and reduce friction [resistance to sliding], making slope failure more likely.

M. Avalanches

Q90. What is an avalanche and what causes it?

An avalanche occurs when snow on a steep mountain slope suddenly becomes unstable and moves downhill. Causes include:

- heavy snowfall in a short period,
 - weak layers inside the snowpack,
 - sudden rise in temperature and partial melting,
 - strong winds piling snow unevenly,
 - earthquakes or vibrations,
 - skiing,
 - trekking, and
 - construction activities.
-

N. Glacial Lake Outburst Floods

Q91. What is a GLOF?

GLOF means **Glacial Lake Outburst Flood**. It happens when a large amount of water stored in a glacial lake is suddenly released.

Q92. How can a GLOF occur?

Rising temperatures may rapidly melt glaciers and increase the water level of glacial lakes. This places pressure on natural dams made of ice or loose moraines. Heavy rainfall or snowfall can add more water. Earthquakes, avalanches or landslides may weaken or damage the dam. If the dam collapses, a large amount of water suddenly rushes downstream and causes destructive flooding.

O. Dust Storms**Q93. What is a dust storm and what causes it?**

A dust storm occurs when strong winds lift large quantities of loose, dry soil and sand into the air. Conditions that increase the risk include:

- prolonged drought,
 - low rainfall,
 - loose dry soil,
 - desert and semi-arid conditions,
 - sparse vegetation,
 - deforestation,
 - overgrazing,
 - poor farming practices,
 - climate change, and
 - extreme weather.
-

CBSE Competency-Based Q&A**Q94. A student observes that rocks are breaking because tree roots have grown into cracks. Is this erosion or weathering? Explain.**

It is **biological weathering**. Plant roots grow into cracks and break rocks apart. The rock is being broken at the same place, so this is weathering rather than erosion.

Q95. A river cuts away soil from one bank and carries it downstream. Which process is taking place?

This is **erosion** because material is not only being worn away but is also being carried from one place to another.

Q96. Why can deforestation increase both erosion and landslide risk?

The chapter directly connects deforestation with landslides because removing vegetation disturbs the natural stability of slopes. It also leaves soil more exposed to erosion. Therefore, deforestation can make the land easier to wear away and can contribute to slope instability.

Q97. Two tectonic plates are moving away from each other and magma is rising between them. Which boundary is this and what may form?

It is a **divergent boundary**. Magma rises and forms new crust. A **mid-ocean ridge** may develop.

Q98. Two continental plates move towards each other. What major landform may be formed? Give the example from the chapter.

Their collision may produce **fold mountains**. The **Himalaya** is the example given in the chapter.

Q99. Why is a delta often suitable for agriculture but also risky for settlements?

A delta contains fertile alluvial soil, so it is suitable for crops such as rice and jute. Rivers also provide transport and fishing opportunities. However, deltas can be **prone to flooding**, so dense settlements may face flood risk.

Q100. Explain the central idea of the chapter in one connected answer.

The Earth's surface is always changing.

Internal forces, such as earthquakes, volcanoes, folding and faulting, move and reshape the crust.

External forces, such as weathering, erosion and deposition, slowly wear down and rebuild landforms.

Running water, waves, glaciers, wind and underground water act as agents of gradation. These processes create different landforms and also influence human life, resources, settlements and cultures. Some landforms are also connected with hazards such as landslides, avalanches, GLOFs and dust storms. Understanding these processes helps people understand the Earth and prepare better for natural disasters.